

il modello lineare

Massimo Borelli

luglio 2025



Massimo Borelli il modello lineare

la fonte storica /1

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Anthropological Miscellanea.

ANTHROPOLOGICAL MISCELLANEA.

REGRESSION *towards* MEDIOCRITY in HEREDITARY STATURE.
By FRANCIS GALTON, F.R.S., &c.

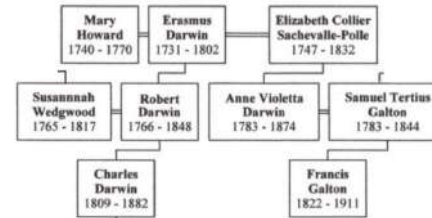
[WITH PLATES IX AND X.]

THIS memoir contains the data upon which the remarks on the Law of Regression were founded, that I made in my Presidential Address to Section H, at Aberdeen. That address, which will appear in due course in the Journal of the British Association, has already been published in "Nature," September 24th. I reproduce here

<https://galton.org/essays/1880-1889/galton-1886-jaigi-regression-stature.pdf>

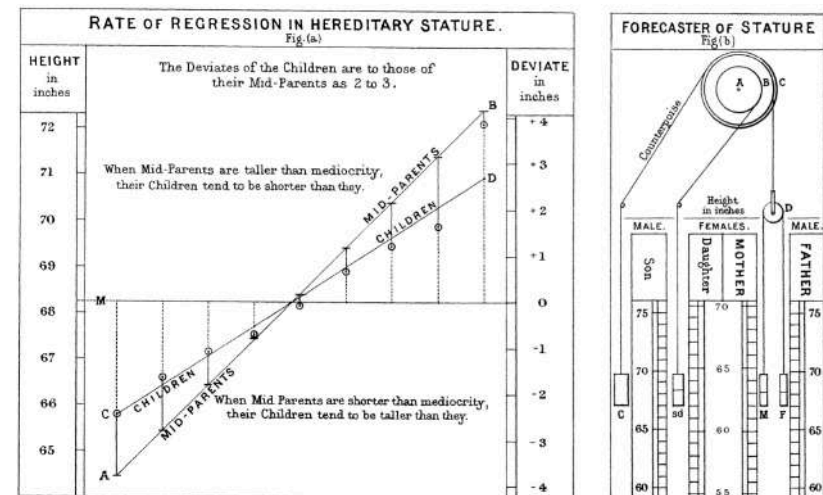
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.. una famiglia mica male!



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la fonte storica /2

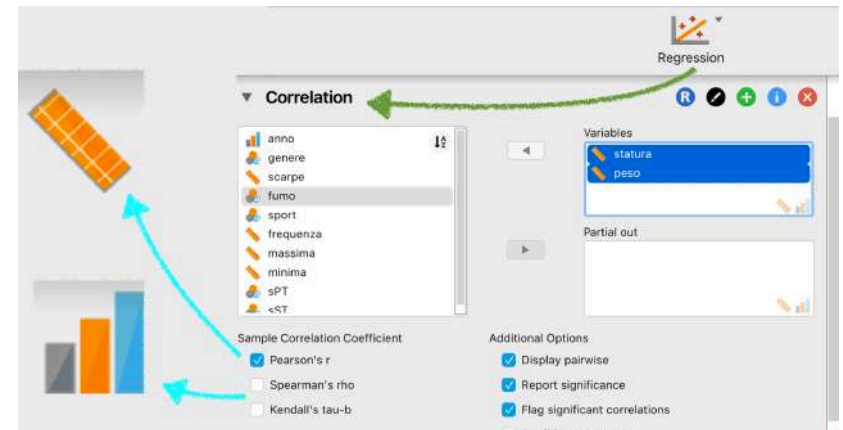


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due strumenti basilari

- la correlazione
- la retta di regressione

la correlazione peso vs. altezza



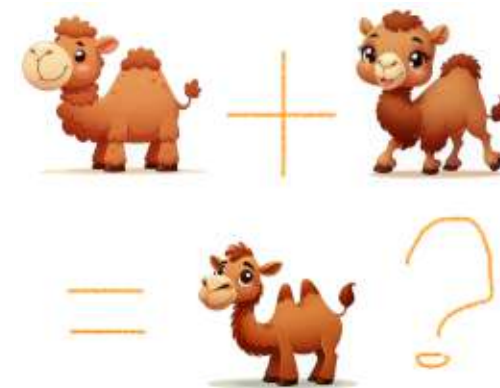
la correlazione peso vs. altezza

Tabella: Pearson's Correlations

		Pearson's r	p
statura	- peso	0.744***	< .001

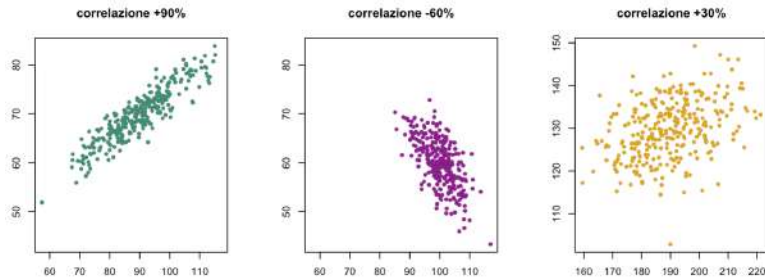
* $p < .05$, ** $p < .01$, *** $p < .001$

vi ricordate?



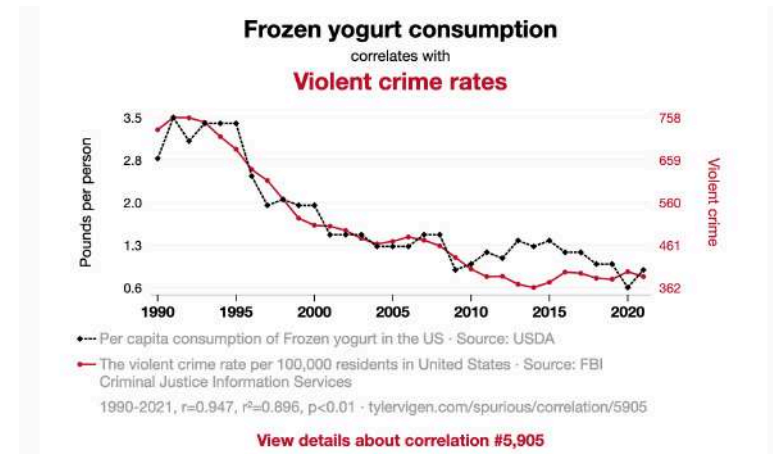
+90%, -60%, +30%

il coefficiente di correlazione ρ di Pearson



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correlation is not causation

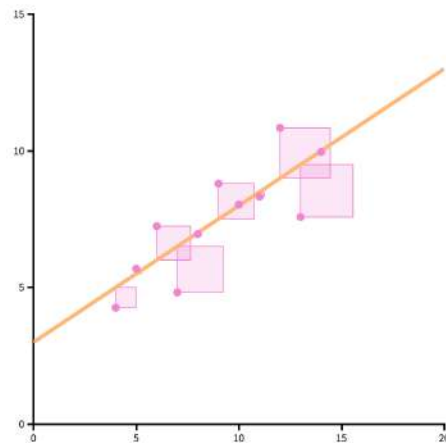


<https://www.tylervigen.com/spurious-correlations>

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il principio dei minimi quadrati

- Galton 1886 ?
- Legendre 1805 ?
- Gauss 1794 ?
- Boskovich 1757.



<https://seeing-theory.brown.edu/>

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la retta di regressione



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Linear Regression

effetti casuali

Model Summary - peso

Model	R	R ²	Adjusted R ²	RMSE
H ₁	0.744	0.554	0.547	6.459

Coefficients

effetti fissi

Model		Unstandardized	Standard Error	t	p
H ₁	(Intercept)	-83.891	16.677	-5.030	< .001
	statura	0.854	0.096	8.850	< .001

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controprova, $p = NS$

Dimmi in che anno sei nata/o e ti dirò quanto pesi ..

Dependent Variable

peso

Method

Enter

Covariates

anno

p

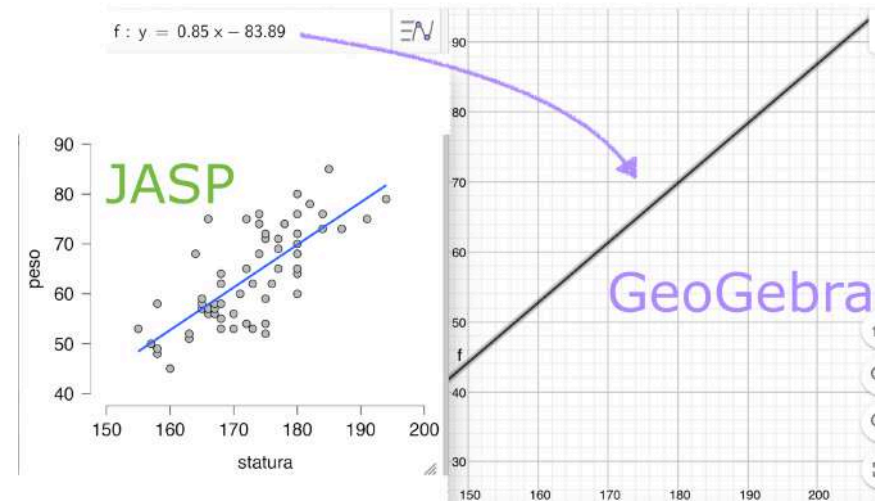
0.935

0.955

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gli effetti fissi

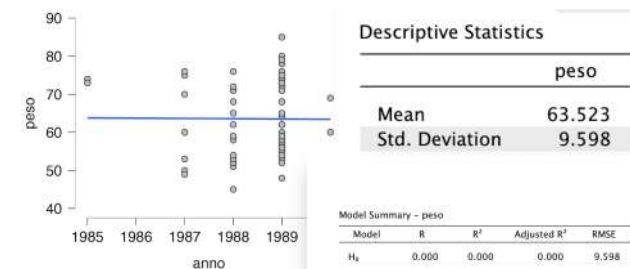


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spieghiamo il modello nullo H₀

Dimmi in che anno sei nata/o e ti dirò quanto pesi ..



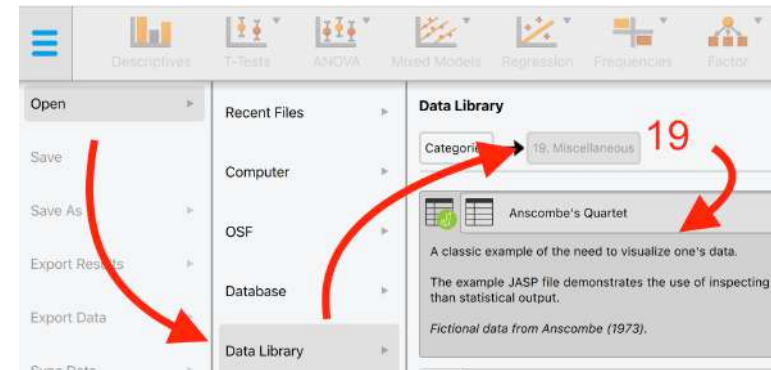
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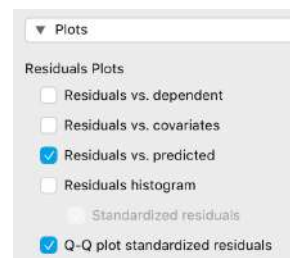
il quartetto di Anscombe

https://it.wikipedia.org/wiki/Quartetto_di_Anscombe

diagnostica della regressione



diagnostica della regressione



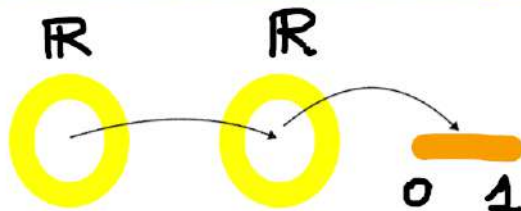
la regressione logistica

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the roma dataset

logHE4	logCA125	logCA19-9	logCEA	AgePatient	Menopause	Histology
3.58	4.25	3.33	0.22	34	ante	benign
3.42	5.45	4.84	0.24	21	ante	benign
5.68	4.72	3.20	0.92	64	post	malignant
4.14	3.96	3.54	1.76	58	post	malignant
3.57	3.03	-0.04	1.03	74	post	benign
3.70	4.11	3.44	0.58	40	ante	benign
7.17	7.58	2.45	0.44	51	ante	malignant
3.57	2.48	1.46	0.10	21	ante	benign
3.97	3.64	2.30	0.14	27	ante	benign
4.11	4.03	4.73	0.82	75	post	malignant
3.56	4.59	4.17	0.34	37	ante	benign
3.65	2.83	3.00	0.71	30	ante	benign
4.89	5.72	2.47	0.44	71	post	malignant



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the roma dataset

Logit

From Wikipedia, the free encyclopedia

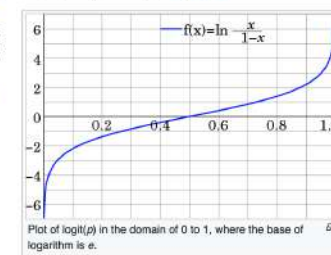
This article discusses the binary logit function only. See *discrete choice* for a discussion of *multinomial logit*, *conditional logit*, *nested logit*, *mixed logit*, *exploded logit*, and *ordered logit*. For the basic regression technique that uses the logit function, see *logistic regression*. For standard magnitudes combined by multiplication, see *logit (unit)*.

In statistics, the **logit** (/ˈloʊdʒɪt/ *LOH-jit*) function is the quantile function associated with the standard logistic distribution. It has many uses in data analysis and machine learning, especially in data transformations.

Mathematically, the logit is the inverse of the standard logistic function $\sigma(x) = 1/(1 + e^{-x})$, so the logit is defined as

$$\text{logit}(p) = \sigma^{-1}(p) = \ln\left(\frac{p}{1-p}\right) \quad \text{for } p \in (0, 1).$$

Because of this, the logit is also called the **log-odds** since it is equal to the **logarithm** of the odds $\frac{p}{1-p}$ where p is a probability. Thus, the logit is a type of function that maps probabilities from $(0, 1)$ to real numbers in $(-\infty, +\infty)$.



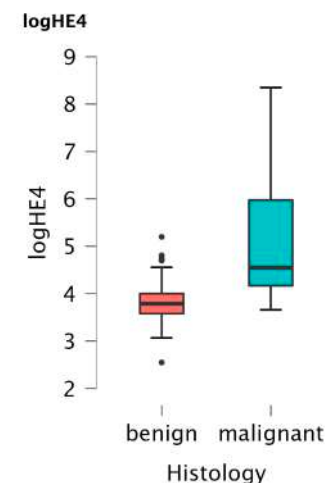
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the roma dataset

The **standard logistic function** is the logistic function with

$$f(x) = \frac{1}{1 + e^{-x}} = \frac{e^x}{e^x + 1} = \frac{1}{2} + \frac{1}{2} \tanh\left(\frac{x}{2}\right).$$

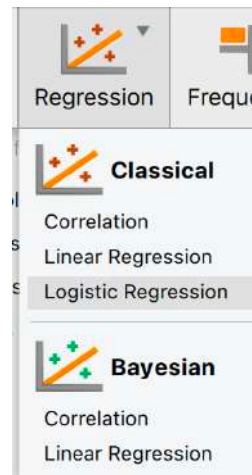
	logHE4	
	benign	malignant
Minimum	2.550	3.660
Maximum	5.200	8.350



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the roma dataset



the roma dataset

	Est.	St. Error	z	Wald Test		
				Wald	df	p
(Intercept)	-14.28	2.38	-6.00	35.98	1	< .001
logHE4	3.07	0.57	5.38	28.94	1	< .001

$$f(x) = -14.28 + 3.07x$$

$$y = \frac{\exp(f(x))}{1 + \exp(f(x))}$$

